

Modelling Software-based Systems

Lecture 4

Validation, Verification and Proof Tools for Event-B for Correct-by-Construction

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- 1 Proof Management
- 2 Development Methodology and Tools for Correct By Construction
- 3 Project Management

Current Summary

- 1 Proof Management
- 2 Development Methodology and Tools for Correct By Construction
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- The panel called *proof control* is providing tools for *discharging* proof obligations
- Proof obligations are stated as follows : $\Gamma \vdash P$: *When assertions of Γ are supposed to hold, then P does hold*
- Problem : Checking that the sequent $\Gamma \vdash P$ is valid.
- How : Applying automatic procedures as pp, pr etc
- ... or helping the proof tool to build the *proof tree*

- Quite sensitive to useless hypotheses.
- Unable to automatically abstract some statements.
- Sometimes poor management of equalities.
- Needs some creative hints

- Proposing **clever lemma** : the user may have interesting ideas because he/she knows the problem to solve.
- Suggesting a **proof by case**.
- Proposing a **witness** for an existential goal.
- Proposing a universal hypothesis **instanciation**.
- etc.

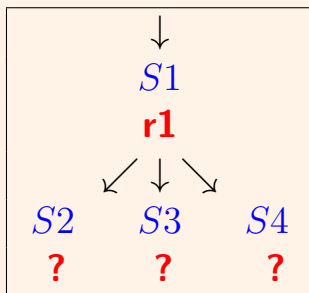
- **Sequent** (where *list_of_hypotheses* might be empty).

$$\textit{list_of_hypotheses} \vdash \textit{conclusion}$$

- **Inference rule** (where *list_of_sequents* might be empty).

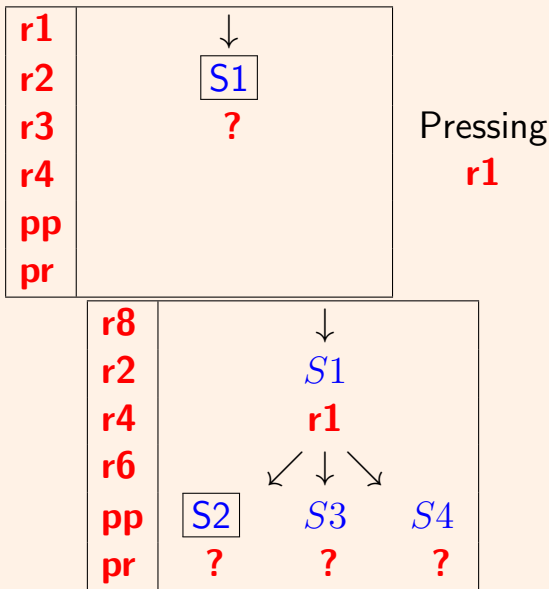
$$\frac{\textit{list_of_sequents}}{\textit{sequent}}$$

- **Backwards application** of inf. rules $\leadsto$ **proof tree**.



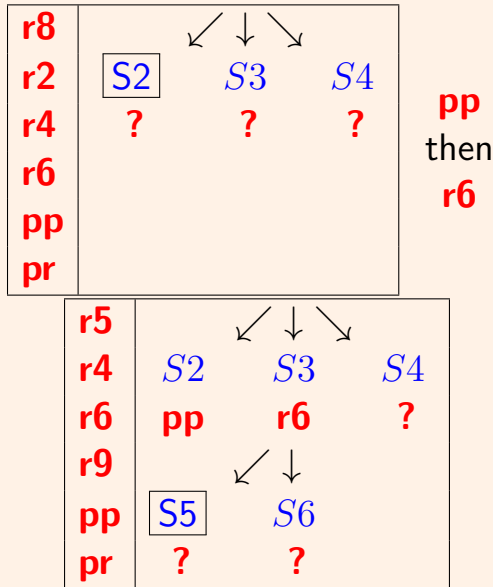
- Rule $r1$ is applied to sequent $S1$.
- Producing sequents $S2$, $S3$, and $S4$.

Interface : Tree and Palette of Rule Buttons



- The proof tree is **constructed interactively**.

Difficulty : Size of Window



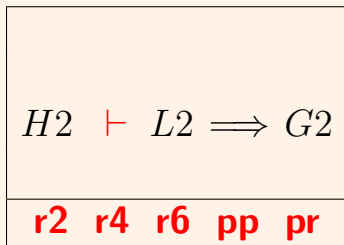
- Our tentative “interface” does not work.

$$H \vdash L \Longrightarrow G$$

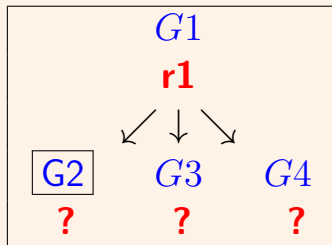
- Most of the time the conclusion has an **implicative form**
- G is called the **goal** (it is a **non-conjunctive** predicate)
- L are called the **local hypotheses** (might be empty)

$$H \vdash I(x) \wedge G(x) \wedge P(x, x') \Longrightarrow I(x')$$

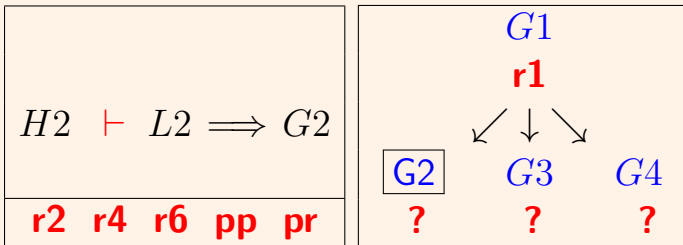
- **Tree window** with **simplified** sequents (**goals** only)
- **Sequent window** containing the **full sequent of interest**



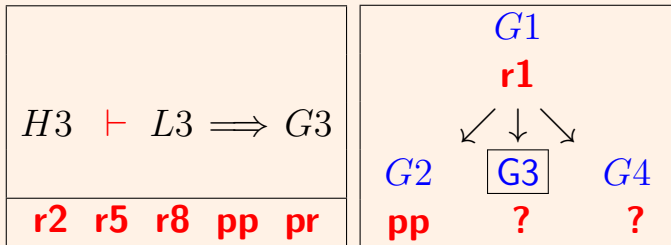
Sequent Window

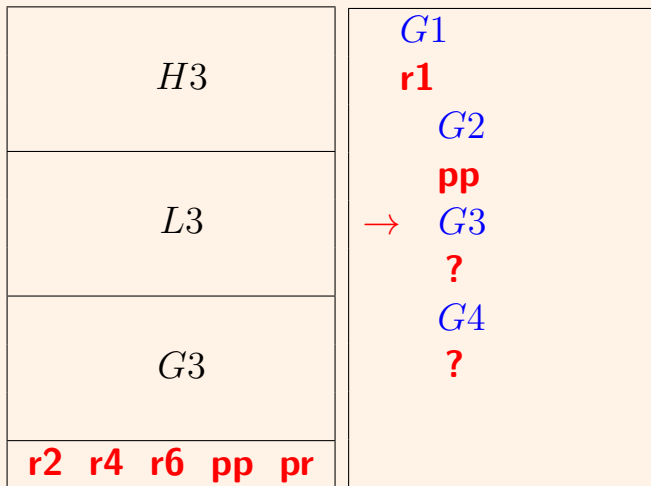


Tree Window



Pressing **pp**





$$\underbrace{hidden ; searched ; cached}_{list_of_hypotheses} \vdash \underbrace{local \implies goal}_{conclusion}$$

$hidden$	Not visible on the sequent window
$searched$	Visible after some search in $hidden$
$cached$	Visible but not part of the conclusion any m
$local$	Visible and part of the conclusion

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Context

Developing life-cycle methodology and a set of associated techniques and tools to develop the highly critical systems using formal techniques from requirements analysis to code generation.

Objectives

- To establish a unified theory for the critical system development.
- To Build a set of tools for supporting new development life-cycle methodology.
- To develop a closed-loop system for verification purpose.
- Graphical based refinement technique to handle the complexity of the system.
- To satisfy requirements and metrics for certifiable assurance and safety.
- To support evidence-based certification.

Development Methodology

- Formal Methods based Critical System Development Life-Cycle

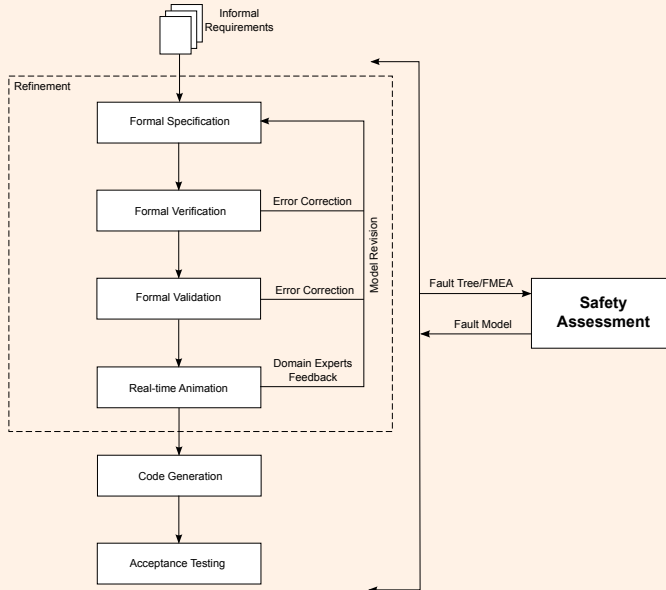
Techniques and Tools

- Real-Time Animator
- Refinement Chart
- Automatic Code Generator : [EB2ALL](#)
- Formal Logic Based Heart-Model

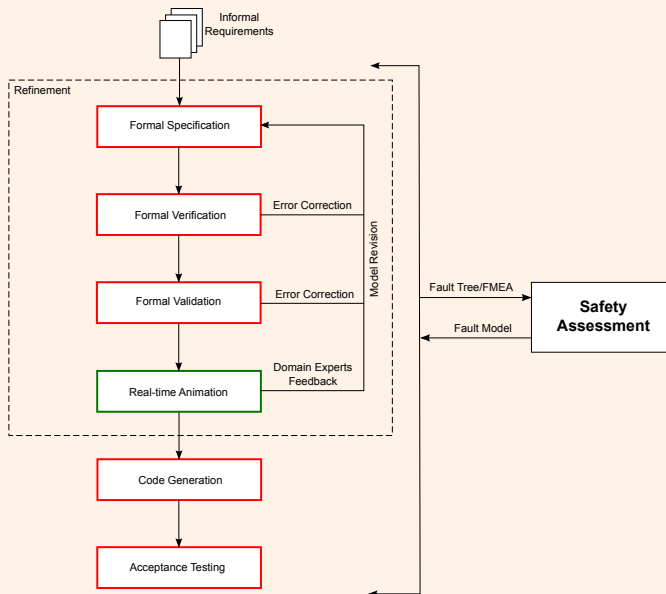
Case Studies

- The Cardiac Pacemaker

Critical System Development Life-Cycle Methodology



Critical System Development Life-Cycle Methodology



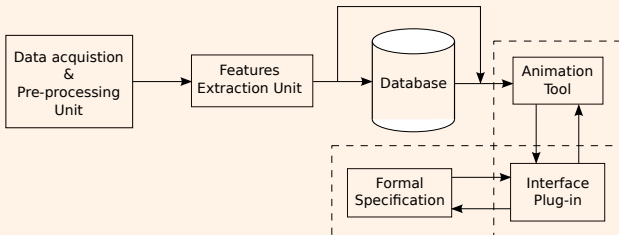
What is Real-time Animator ?

- Visual representation of formal model using real time data set.

Why should we use Formal Model Animator ?

- To validate system behavior according to the stakeholders
- To express formal models for non-mathematical domain experts
- To discover the error in the early stage of system development (Traceability)

Proposed Architecture

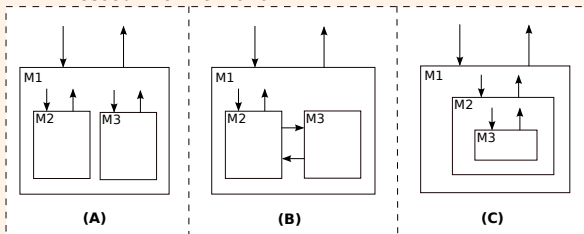


- Refinement chart is a graphical representation of a complex system using layering approach.
- Integration and features based code structuring of different components of the systems.

Basic rules of refinement chart

- Parallel Refinement
- Sequential Refinement
- Nested Refinement

$$\begin{aligned} M1 &\sqsubseteq (M2 \parallel M3 \parallel \dots \parallel M_{n-1} \parallel M_n) \\ M1 &\sqsubseteq (M2 \succ M3 \succ \dots \succ M_{n-1} \succ M_n) \\ (M1 &\sqsubseteq M2, M2 \sqsubseteq M3, \dots, M_{n-1} \sqsubseteq M_n) \end{aligned}$$



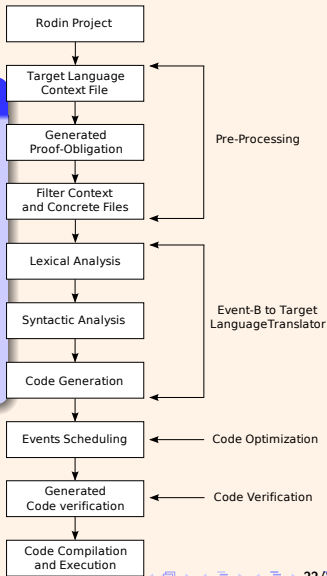
$$\begin{aligned} M1 &\sqsubseteq (M2 \parallel M3) \\ M1 &\sqsubseteq (M2 \succ M3) \\ (M1 &\sqsubseteq M2, M2 \sqsubseteq M3) \end{aligned}$$

EB2ALL

- Automatic code generation from Event-B formal specification to multiple languages (C, C++, Java, C#) to make it executable.
- A module for Rodin (Deploy is an European Commission FP7 project)
- It is more than 20000 lines of code
- More than 1000 downloads

Download : <http://eb2all.loria.fr/>

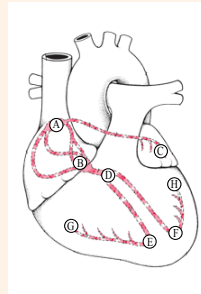
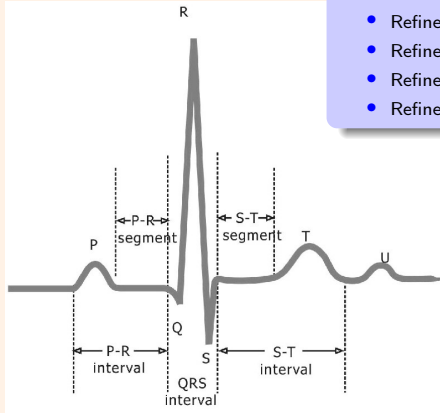
EB2ALL Tool Architecture



Formal Development

- Abstract Model
- Refinement 1 : Introducing Steps in the Propagation
- Refinement 2 : Impulse Propagation
- Refinement 3 : Perturbation the Conduction
- Refinement 4 : Getting a Cellular Model

Electrocardiogram (ECG)



Landmarks on the Electrical Conduction of Heart

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- **Property**

- ▶ the set of people is non empty and finite
- ▶ $people \neq \emptyset \wedge finite(people)$